26900186 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECT300, MG5DSCECC300 - DIGITAL DESIGN USING VERILOG

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. In the Verilog number 12'd25, what does d indicate? [K] / [CO1]

   a). Double b). Decimal

   c). Data d). Digit

2. Which gate should combine the individual product terms of a 4-to-1 multiplexer? [A] / [CO2]

   a). AND b). NOT

   c). NOR d). OR

3. Which Verilog statement is used to select one option from multiple conditions? [K] / [CO3]

   a). case b). always c). wire d). reg

4. module AND_gate(input a, b, output y); assign y = a & b; endmodule, In this code which keyword represents continous assignment statement [U] / [CO1]

   a). module b). Input

c). assign d). endmodule

5. Which Verilog expression correctly implements the Sum of a Full Adder? [U] / [CO2]

a). assign Sum = A & B & Cin; b). assign Sum = A | B | Cin; c). assign Sum = A ^ B ^ Cin; d). assign Sum = ~ (A ^ B ^ Cin);

6. Which modeling level provides the closest representation of the actual logical gates used in a digital circuit? [U] / [CO1]

a). Behavioral b). Dataflow

c). Gate d). Algorithmic

7. What is the purpose of reset in an asynchronous counter? [A] / [CO2]

a). Increase the counting speed b). Force the counter to a known initial state

c). Generate the clock d). Disable all gates permanently

8. Explain the role of a flip-flop in digital systems. [U] / [CO3]

a). It only performs signal amplification b). It stores a binary state and can be used as a basic memory element

c). It converts every input into multiple outputs d). It permanently removes clock signals

9. What does the operator ~ perform? [K] / [CO1]

a). Logical NOT b). Bitwise NOT

c). Logical AND d). XOR

10. Calculate the simulation time after executing '#10 a = b;' when the statement begins at 20 ns. [A] / [CO3]

a). 30 ns b). 10 ns

c). 20 ns d). 200 ns

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. What is Verilog and why is it called a Hardware Description Language? [K] / [CO1]

12. Explain the operation of a 3-to-8 decoder. Derive the Boolean expression for each output. [U] / [CO2]

13. Explain the main components of a Verilog testbench. [K] / [CO3]

14. What is meant by concurrency in Verilog? How is it different from sequential execution in a conventional programming language? [U] / [CO1]

15. Obtain the data flow expressions for XOR and XNOR gates. [U] / [CO2]

16. Explain the operation of a 4-bit shift register. Discuss serial-in serial-out (SISO) with suitable Verilog statements. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the operation of a Full Subtractor. Design and implement a Full Subtractor using gate-level modeling with wire elements. Write a suitable testbench to verify its operation. [A] / [CO2]

18. Explain the basic structure of a Verilog program. Describe the purpose of modules, ports, declarations, and statements [U] / [CO1]

19. From the following Verilog declarations and answer the questions that follow: wire [7:0] data; reg [3:0] count; integer i; real voltage; time delay; (a) Identify the data type of each declaration. (b) State the number of bits specified for data and count. (c) Mention the purpose of integer, real, and time. (d) Explain why different data types are required in Verilog HDL. (e) Write suitable Verilog declarations for a 16-bit signal and an 8-bit procedural variable. [U] / [CO1]

20. Explain the design and behavioral modeling of a 2-to-4 decoder. Include its block diagram description, truth table, Verilog code and applications. [U] / [CO3]

[10x2 = 20]